

## KERALA DIPLOMA CURRICULUM REVISION 2026

Course 5092

5 Credits: 4

Program Core

Source-grounded

# Quality Assurance & Testing

□ To understand the significance in quality control of polymer products.

## OFFICIAL SOURCE DECLARATION

### How this lesson was prepared

The supplied official SITTTTR syllabus PDF for Course 5092 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

**Source file:** 5092.pdf from the uploaded official SITTTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

| Item                 | Official information                                 |
|----------------------|------------------------------------------------------|
| Programme            | Diploma in Polymer Technology                        |
| Course code          | 5092                                                 |
| Course title         | Quality Assurance &Testing                           |
| Semester             | 5 Credits: 4                                         |
| Course category      | Program Core                                         |
| Periods per week     | 4 (L:3, T:1, P:0) Periods per semester: 60           |
| Periods per semester | 60                                                   |
| Credits              | 4                                                    |
| Prerequisite         | See the official syllabus PDF                        |
| Assessment           | Use the official assessment section reproduced below |

Modules

4 official module sections detected.

Topic entries

16 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

**STUDY METHOD**

## How to use this lesson

### First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

### Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

### Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

### Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

**LEARNING OUTCOMES**

# Objectives, outcomes and mapping

## Course objectives

- To understand the significance in quality control of polymer products.
- To impart knowledge on raw material testing, processability tests and product testing
- which comes under both rubbers and plastics.

## Course outcomes

| CO  | Official outcome                                                                                          | Study level           |
|-----|-----------------------------------------------------------------------------------------------------------|-----------------------|
| CO1 | 15 Understanding and, electrical properties of polymers. Summarize test methods for ISNR and concentrated | See official syllabus |
| CO2 | 14 Understanding latex as per IS standards Explain the testing procedures for physical                    | See official syllabus |
| CO3 | 15 Understanding properties of polymers. Explain the specific testing procedures for dry                  | See official syllabus |
| CO4 | 14 Understanding rubber, plastic and latex products                                                       | See official syllabus |

## CO-PO mapping evidence

| Row | Extracted official mapping |
|-----|----------------------------|
| CO1 | CO1 3                      |
| CO2 | CO2 3                      |
| CO3 | CO3 3                      |
| CO4 | CO4 3                      |

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

| Module   | Official heading                                                   | Topic entries | Hours        |
|----------|--------------------------------------------------------------------|---------------|--------------|
| Module 1 | properties of polymers                                             | 4             | As specified |
| Module 2 | standards                                                          | 4             | As specified |
| Module 3 | Explain the testing procedures for physical properties of polymers | 4             | As specified |
| Module 4 | products                                                           | 4             | As specified |

**MODULE 1**

# properties of polymers

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: properties of polymers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ 1 Remembering of standards organizations Summarize the role of standards

1 Remembering of standards organizations Summarize the role of standards is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Remembering of standards organizations Summarize the role of standards using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 remembering of standards organizations summarize the role of standards should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                    |
|---------------------|--------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 1 Remembering of standards organizations Summarize the role of standards means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1- 1 Remembering of standards organizations Summarize the role of standards definition/meaning . . . , steps, application . . . . Technical terms English- . . . .

## ▼ organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw

organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, organizations in standardization and quality 3 understanding assurance. explain test procedures for purity of raw should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What organizations in standardization and quality 3 Understanding assurance. Explain test procedures for purity of raw means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-ഓർഗനൈസേഷനുകളുടെ സ്റ്റാൻഡറൈസേഷൻ ആൻഡ് ക്വാളിറ്റി അസ്യൂറൻസ് 3 Understanding assurance. Explain test procedures for purity of raw ഓർഗനൈസേഷനുകളുടെ സ്റ്റാൻഡറൈസേഷൻ ആൻഡ് ക്വാളിറ്റി അസ്യൂറൻസ്. ഓർഗനൈസേഷനുകളുടെ സ്റ്റാൻഡറൈസേഷൻ ആൻഡ് ക്വാളിറ്റി അസ്യൂറൻസ്, steps, application ഓർഗനൈസേഷനുകളുടെ സ്റ്റാൻഡറൈസേഷൻ ആൻഡ് ക്വാളിറ്റി അസ്യൂറൻസ്. Technical terms English-ഓർഗനൈസേഷൻ സ്റ്റാൻഡറൈസേഷൻ ആൻഡ് ക്വാളിറ്റി അസ്യൂറൻസ്.

## ▼ 4 Understanding materials Describe test methods for thermal and

4 Understanding materials Describe test methods for thermal and is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding materials Describe test methods for thermal and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding materials describe test methods for thermal and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding materials Describe test methods for thermal and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-4 Understanding materials Describe test methods for thermal and definition/meaning. steps, application. Technical terms English-.

▼ **7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS**

7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS is an official topic in Module 1 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

## Study points

- Define or identify 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 7 understanding electrical properties of polymers. importance of quality assurance and testing-quality-definition. modern quality concepts, importance of quality control in polymer industry. standards and specifications. different standard organizations - iso, astm, bis, din. procedure for accreditation of isi certification. role of standard organization in quality assurance and customer satisfaction. raw material testing - testing of additives-% purity of dpg, mbt, zno. thermal properties - thermal conductivity, flammability, vicat softening temperature, melting point, brittleness temperature, specific heat, hdt. electrical properties - dielectric strength, dielectric constant, arc resistance. summarize test methods for isnr and concentrated latex as per is should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

## Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 1-7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS

definition/meaning പദപരിഭാഷണം. പദപരിഭാഷണം പദപരിഭാഷണം, steps, application പദപരിഭാഷണം പദപരിഭാഷണം പദപരിഭാഷണം. Technical terms English-പദപരിഭാഷണം പദപരിഭാഷണം.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 2**

# standards

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: standards
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR

procedure to determine tests of dry rubber 6 Understanding (ISNR).  
Compile the technical specifications of ISNR is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, procedure to determine tests of dry rubber 6 understanding (isnr). compile the technical specifications of isnr should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                           |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-എന്നുവേണ്ടി dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR എന്നതിനുള്ള definition/meaning എഴുതുക. എന്നതിനുള്ള steps, application എന്നിവ എഴുതുക. Technical terms English-ൽ എഴുതുക.

## ▼ 1 Understanding as per IS 4588. Explain the significance, principle and

1 Understanding as per IS 4588. Explain the significance, principle and is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding as per IS 4588. Explain the significance, principle and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 1 understanding as per is 4588. explain the significance, principle and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 1 Understanding as per IS 4588. Explain the significance, principle and means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                           |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                 |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- 1 Understanding as per IS 4588. Explain the significance, principle and definition/meaning. steps, application. Technical terms English- .

## ▼ procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of

procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, procedure to determine tests of concentrated 6 understanding latex. compile the technical specifications of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                       |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                               |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                     |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2- $\square\square$  procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of  $\square\square\square\square\square\square\square\square\square$   $\square\square\square\square$  definition/meaning  $\square\square\square\square\square\square\square\square\square$ .  $\square\square\square\square$   $\square\square\square\square$   $\square\square\square\square\square\square$ , steps, application  $\square\square\square\square$   $\square\square\square\square\square\square\square$   $\square\square\square\square$ . Technical terms English- $\square$   $\square\square\square\square$   $\square\square\square\square\square\square\square\square$ .

**▼ 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity**

1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex – TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity is an official topic in Module 2 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### **Study points**

- Define or identify 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content,

Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 1 understanding concentrated latex as per IS 3708. significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N<sub>2</sub> content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for viscosity and specific gravity should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N <sub>2</sub> content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex –TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 2-1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR – dirt content, volatile matter, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex – TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity definition/meaning, steps, application. Technical terms English-Malayalam.

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 3**

# Explain the testing procedures for physical properties of polymers

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

## Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

## Key facts

- Official module title: Explain the testing procedures for physical properties of polymers
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ 4 Understanding properties of plastics and rubbers Describe the test method to determine

4 Understanding properties of plastics and rubbers Describe the test method to determine is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding properties of plastics and rubbers Describe the test method to determine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding properties of plastics and rubbers describe the test method to determine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                    |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding properties of plastics and rubbers Describe the test method to determine means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-4 Understanding properties of plastics and rubbers Describe the test method to determine definition/meaning. steps, application. Technical terms English-  
.

## ▼ 4 Understanding permanent set of rubbers. Describe the test method to determine

4 Understanding permanent set of rubbers. Describe the test method to determine is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding permanent set of rubbers. Describe the test method to determine using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding permanent set of rubbers. describe the test method to determine should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                           |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding permanent set of rubbers. Describe the test method to determine means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                   |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                         |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3-4 Understanding permanent set of rubbers. Describe the test method to determine definition/meaning. ഏതു ഏതു ഏതു, steps, application ഏതു ഏതു. Technical terms English-ഏതു ഏതു.

## ▼ hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of

hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, hardness, abrasion resistance, resilience, flex 6 understanding crack, fatigue failure and impact testing. outline the procedure to determine mfi of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.



▼ **1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex**

1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex is an official topic in Module 3 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics.

Explain the specific testing procedures for dry rubber, plastic and latex using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

## Engineering or professional relevance

In Polytechnic practice, 1 understanding plastics. method of testing, principle of conditioning. significance, specimen standards, test methods relevant to plastics and rubbers – tensile strength, m300 , elongation at break, tear strength, tension set, compression set, hardness, abrasion resistance, resilience, flex cracking, fatigue failure, impact testing of plastics- izod and charpy processability tests - determination of melt flow index of plastics. explain the specific testing procedures for dry rubber, plastic and latex should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers – Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 3- 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers – Tensile strength, M300 , Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests -Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex   
 definition/meaning .  
 steps, application   
 Technical terms English-   
 .

► **Official syllabus detail and terminology (expand in digital version)**

**Module study routine:** Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

**MODULE 4****products**

This module is presented from the official syllabus scope for Quality Assurance & Testing. Use the topic cards for learning and the source transcript for exact coverage.

**As specified**  
official hours

**CO-linked**  
see outcomes

**Learning goals**

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

**Key facts**

- Official module title: products
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

## ▼ State the importance of product testing. 1 Remembering Outline the test procedure and specifications

State the importance of product testing. 1 Remembering Outline the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify State the importance of product testing. 1 Remembering Outline the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, state the importance of product testing. 1 remembering outline the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What State the importance of product testing. 1 Remembering Outline the test procedure and specifications means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                        |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                              |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- State the importance of product testing. 1 Remembering Outline the test procedure and specifications definition/meaning . steps, application . Technical terms English- .

## ▼ 5 Understanding for dry rubber products. Explain the test procedure and specifications

5 Understanding for dry rubber products. Explain the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 5 Understanding for dry rubber products. Explain the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 5 understanding for dry rubber products. explain the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                                  |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 5 Understanding for dry rubber products. Explain the test procedure and specifications means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                          |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-5 Understanding for dry rubber products. Explain the test procedure and specifications definition/meaning. steps, application. Technical terms English- Malayalam.

## ▼ 4 Understanding for latex products. Explain the test procedure and specifications

4 Understanding for latex products. Explain the test procedure and specifications is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding for latex products. Explain the test procedure and specifications using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding for latex products. explain the test procedure and specifications should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect | What to prepare |
|--------|-----------------|
|--------|-----------------|

| Aspect              | What to prepare                                                                                                             |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding for latex products. Explain the test procedure and specifications means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                     |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                           |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4- 4 Understanding for latex products. Explain the test procedure and specifications definition/meaning. , steps, application . Technical terms English- .

**▼ 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com>**

4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> is an official topic in Module 4 of Quality Assurance & Testing. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

### **Concept and explanation**

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official

source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

### Study points

- Define or identify 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers – Scott R1 Physical testing of Rubbers – R.P. Brown R2 Indian Standards R3 ASTM standards – Vol. 8 & 9 series R4 Plastic Testing Technology – Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

### Engineering or professional relevance

In Polytechnic practice, 4 understanding for plastic products. importance of product testing, testing of dry rubber products- is specifications and specific test procedures of cycle tyre, cycle tube, m.c. sheet, v-straps. testing of latex products- is specifications and specific test procedures of latex foam, surgical gloves and examination gloves. testing of plastic product - pvc pipe, water tank, helmet text / reference: t/r book title/author t1 physical testing of rubbers – scott r1 physical testing of rubbers – r.p. brown r2 indian standards r3 astm standards – vol. 8 & 9 series r4 plastic testing technology – vishu shah online resources: sl.no website link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com> should be discussed with attention to safe

operation, correct terminology, observable evidence and the relationship between input, process and result.

### Topic study guide

| Aspect              | What to prepare                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition/identity | What 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers – Scott R1 Physical testing of Rubbers – R.P. Brown R2 Indian Standards R3 ASTM standards – Vol. 8 & 9 series R4 Plastic Testing Technology – Vishu Shah Online Resources: SI.No Website Link 1 <a href="https://www.astm.org">https://www.astm.org</a> 2 <a href="https://www.astm.org">https://www.astm.org</a> 3 <a href="https://link.springer.com">https://link.springer.com</a> 4 <a href="http://www.intertek.com">http://www.intertek.com</a> 5 <a href="https://www.smithersrapra.com">https://www.smithersrapra.com</a> means in the official course context |
| Study lens          | Principle → components or stages → result → application                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Exam evidence       | Definition, labelled explanation, comparison, procedure or example as appropriate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

**Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

**Common mistakes:** Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

**Malayalam support:** Module 4-□□ 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers – Scott R1 Physical testing of



**RAPID REVISION****Diagram and source-transcript library****Module 1 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 2 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 3 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**Module 4 source and topic cards**

Jump to official terminology, topic explanations and study guidance.

**OFFICIAL SELF-LEARNING****Activities from the syllabus**

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

## EXAMINATION PREPARATION

# Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

**Practice-paper notice:** The model paper below is generated for revision and is not an official examination paper.

**ACTIVE RECALL**

## Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

### Module 1

► **Define or explain 1 Remembering of standards organizations**  
**Summarize the role of standards. (3 marks)**

► **Define or explain organizations in standardization and quality 3**  
**Understanding assurance. Explain test procedures for purity of raw. (3 marks)**

► **Define or explain 4 Understanding materials Describe test methods for thermal and. (3 marks)**

► **Define or explain 7 Understanding electrical properties of polymers. Importance of quality assurance and Testing-Quality-definition. Modern quality concepts, importance of quality control in polymer industry. Standards and specifications. Different standard organizations - ISO, ASTM, BIS, DIN. Procedure for accreditation of ISI certification. Role of standard organization in quality assurance and customer satisfaction. Raw material testing - Testing of Additives-% Purity of DPG, MBT, ZnO. Thermal properties - Thermal conductivity, flammability, Vicat softening temperature, Melting point, brittleness temperature, specific heat, HDT. Electrical properties - Dielectric strength, dielectric constant, arc resistance. Summarize test methods for ISNR and concentrated latex as per IS. (3 marks)**

### Module 2

► **Define or explain procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR. (3 marks)**

► **Define or explain 1 Understanding as per IS 4588. Explain the significance, principle and. (3 marks)**

► **Define or explain procedure to determine tests of concentrated 6 Understanding latex. Compile the technical specifications of. (3 marks)**

► **Define or explain 1 Understanding concentrated latex as per IS 3708. Significance, principle and method of tests for ISNR - dirt content, volatile mater, ash content, N2 content, Po and PRI, ASH and Colour. Technical specifications of ISNR as per BIS. Significance, principle and methods of tests for Concentrated latex -TSC, DRC, KOH number, VFA number, Mechanical stability time, Mg content, Sludge and Coagulum content, Alkalinity etc. Technical specifications for concentrated latex as per BIS. Test methods for Viscosity and Specific gravity. (3 marks)**

## **Module 3**

► **Define or explain 4 Understanding properties of plastics and rubbers Describe the test method to determine. (3 marks)**

► **Define or explain 4 Understanding permanent set of rubbers. Describe the test method to determine. (3 marks)**

► **Define or explain hardness, abrasion resistance, resilience, flex 6 Understanding crack, fatigue failure and impact testing. Outline the procedure to determine MFI of. (3 marks)**

► **Define or explain 1 Understanding plastics. Method of testing, principle of conditioning. Significance, specimen standards, test methods relevant to plastics and rubbers - Tensile strength, M300 ,**

**Elongation at break, Tear strength, Tension set, Compression set, Hardness, Abrasion resistance, Resilience, Flex cracking, Fatigue failure, Impact testing of Plastics- Izod and Charpy Processability tests - Determination of melt flow index of plastics. Explain the specific testing procedures for dry rubber, plastic and latex. (3 marks)**

## **Module 4**

► **Define or explain State the importance of product testing. 1 Remembering Outline the test procedure and specifications. (3 marks)**

► **Define or explain 5 Understanding for dry rubber products. Explain the test procedure and specifications. (3 marks)**

► **Define or explain 4 Understanding for latex products. Explain the test procedure and specifications. (3 marks)**

► **Define or explain 4 Understanding for plastic products. Importance of product testing, testing of dry rubber products- IS specifications and specific test procedures of Cycle tyre, Cycle tube, M.C. sheet, V-straps. Testing of latex products- IS specifications and specific test procedures of Latex foam, Surgical gloves and Examination gloves. Testing of Plastic product - PVC pipe, Water tank, Helmet Text / Reference: T/R Book Title/Author T1 Physical testing of Rubbers - Scott R1 Physical testing of Rubbers - R.P. Brown R2 Indian Standards R3 ASTM standards - Vol. 8 & 9 series R4 Plastic Testing Technology - Vishu Shah Online Resources: Sl.No Website Link 1 <https://www.astm.org> 2 <https://www.astm.org> 3 <https://link.springer.com> 4 <http://www.intertek.com> 5 <https://www.smithersrapra.com>. (3 marks)**

**PRACTICE ONLY**

## Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

### Module 1

1-mark definition: State the meaning of **1 Remembering of standards organizations Summarize the role of standards.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 2

1-mark definition: State the meaning of **procedure to determine tests of dry rubber 6 Understanding (ISNR). Compile the technical specifications of ISNR.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

### Module 3

1-mark definition: State the meaning of **4 Understanding properties of plastics and rubbers Describe the test method to determine.**

### Module 4

1-mark definition: State the meaning of **State the importance of product testing. 1 Remembering Outline the test procedure and specifications.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

## QUICK REVISION

### Exam checklist

#### Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

#### Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

**REFERENCE VOCABULARY**

# Glossary

**Study glossary**

| Term              | Meaning in this lesson                                                   |
|-------------------|--------------------------------------------------------------------------|
| Course Outcome    | A capability the student should demonstrate after completing the course. |
| Module            | An official syllabus unit with defined topics and hours.                 |
| CIE               | Continuous Internal Evaluation.                                          |
| ESE               | End Semester Examination.                                                |
| Source transcript | A preserved extract of the official syllabus used for coverage checking. |

**SOURCE-SUPPORTED REFERENCES**

## References and web resources

**References listed in the official PDF**

References are included in the official source PDF.

**Web resources listed in the official PDF**

- Sl.No Website Link <https://www.astm.org> <https://www.astm.org>  
<https://link.springer.com> <http://www.intertek.com> <https://www.smithersrapra.com>

**IN-PAGE SEARCH****Search this lesson**

Prepared from the supplied official SITTTTR syllabus PDF for Course 5092. Verify institutional updates against the current official curriculum.